Can we trust the machine? Utility and accuracy of wildlife classification models for locating rare species in a metropolitan camera trapping network

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Background and Objectives

Cleveland Metroparks (CMP) began its large-scale camera trapping project—Focus on Wildlife[1]—in 2015. Currently, the project covers up to 200+ camera sites at once for continual wildlife monitoring, making it one of the largest and longest running camera trapping projects in the country. With over 650k camera-nights of data and 28 million total images collected across 10 years, a manual re-evaluation of all image data would be nearly impossible.

Despite this, validation of species occurrences is necessary to ensure optimal management of wildlife at CMP, especially as the tools, technology, and scale of the project have evolved. For this reason, we employ SpeciesNet[2], a species classification computer vision model, to assist with (1) targeted validation of existing rare species observations and (2) assistance with locating images that may be rare species but were not tagged as such upon first review. Species considered rare for CMP's camera network include: **gray fox, bobcat, black bear, flying squirrel, and all mustelids (weasel, mink, otter, badger, etc.)**.

Methods

We focused our efforts on images from the CMP catalog that were tagged by novice observers before SpeciesNet was available and fell within the Urban Wildlife Information Network[3] survey months (January, April, July, October). These images are overall more likely to have species misidentifications and have high value for CMP's own wildlife monitoring efforts and for research partners.

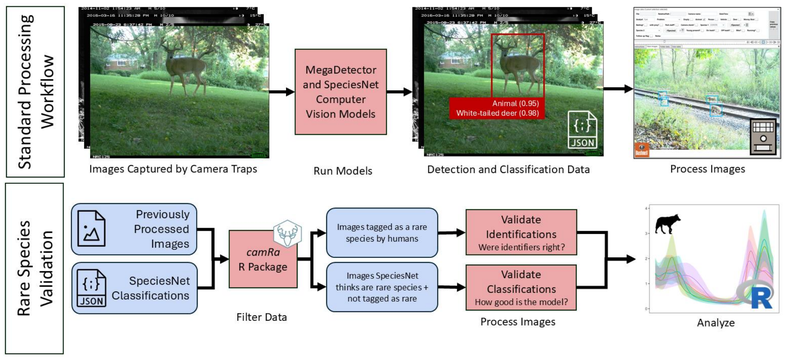


Figure 1. Simplified image processing workflows with SpeciesNet. Confidence thresholds for MegaDetector and SpeciesNet are used for standard processing, but not for rare species validation.

Standard Processing

- Images are processed through MegaDetector[4] (and now SpeciesNet[4]) to filter out images that are likely blank.
- Humans review every image that is tagged by hand using Timelapse[5], now granular filtering by predicted species.

Rare Species Validation

- Images tagged as a rare species by a human or predicted to be a rare species by SpeciesNet were filtered out to re-assessed.
- Existing IDs for each image and SpeciesNet classifications were compared against validated IDs to determine accuracy for both SpeciesNet and observers.

Results

We reviewed a total of 2,857 images classified as a rare species by either humans or SpeciesNet. 42% (1,192) of these images contained a rare species (Fig. 3, left side) while the remaining 58% (1,665) of images were common species, undifferentiable (unknown) wildlife, or empty images (Fig. 3, right side). False positives were greatly inflated by the bear classification within SpeciesNet, as these were mostly images of animals that could not be differentiated (Unknown; Fig 3, Fig. 4).

SpeciesNet did notably increase the number of images for all three rare species observed in the dataset by 9%, 21%, and 39%, respectively. We did not observe gray foxes, bobcats, black bears, or otters, despite some images originally being tagged as these species.

Identifications by humans were mostly accurate with only 4% of images being classified as a rare species erroneously, in range of error rates seen in other datasets[6]. Confusion here was mostly with other species that were morphologically similar, such as cats for bobcats and squirrels for weasels and minks (examples in Fig. 2; Fig. 4).



Figure 2. False positive images tagged by SpeciesNet that appear to be correct at first glance and may confuse novice observers.

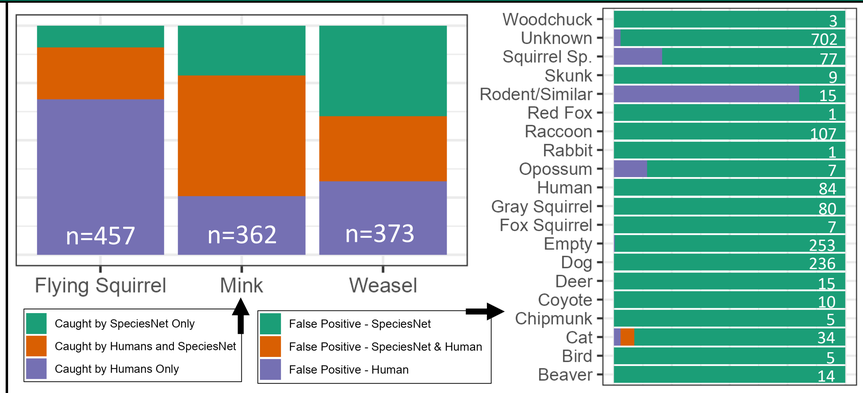


Figure 3. Totals and proportions of rare species images (true positives) and images of other species (false positives) tagged as a rare species by how they were located (IDed by a human or SpeciesNet).

Figure 4. Sankey diagrams of original IDs compared to validated IDs for images tagged as rare by humans and SpeciesNet. Some images appear twice (once left and right). "Other Taxa" includes all non-rare species.

So, can we trust the machine?

SpeciesNet can be a highly effective tool for enhancing image processing workflows, both by increasing processing speeds and allowing for fast validation of high-value camera data. For rare species, error rates are still high, and we do not recommend using SpeciesNet to independently tag images. For some common species with more training data (e.g., deer and coyote), error rates have been shown to be low enough that it can be used alone for certain analyses[7].

Despite its high error rates here, SpeciesNet, did increase the total number of images for all three species found by providing a "different set of eyes" on the data. As more data are incorporated into the model, error rates are likely to decline as SpeciesNet to becomes more robust, which may allow for some level of automated image tagging.